

A Summary of “Shannon Entropy and Kolmogorov Complexity”

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1 Introduction

Information Theory is concerned with the question, “How can we communicate a message using the least number of bits?”. Claude Shannon posed and answered this question in his seminal 1948 paper, devising encoding schemes based on the frequency of characters. Shannon information theory quantifies the information content of a message, modeled as a series of outcomes of a random variable, using *entropy*. Algorithmic information theory, developed concurrently by Ray Solomonoff, Andrey Kolmogorov, and Gregory Chaitin, is concerned with the information inherent to objects represented as binary strings, irrespective of any probability distribution.

Both Shannon information theory and algorithmic information theory quantify information using *bits*, but do so using very different language from probability theory and theory of computation. In their survey paper titled “Shannon Entropy and Kolmogorov Complexity,” Grünwald and Vitányi relate analogous concepts in Shannon information theory (entropy, mutual information, sufficient statistics, rate distortion) and algorithmic information theory (Kolmogorov complexity, algorithmic mutual information, algorithmic statistics, structure function) using expectation [1].

This summary covers the main theorems of their paper. Key terms are defined as needed and with emphasis on intuition rather than rigor. Supplemental explanations from Cover and Thomas’s classical textbook *The Elements of Information Theory* are included whenever helpful [2].

2 Main Results

2.1 Shannon Entropy and Kolmogorov Complexity

Shannon entropy $H(X)$ is the average amount of information obtained from observing the output of random variable X . Entropy quantifies the uncertainty in X . More practically, entropy is interpreted as the length (number of bits) needed to encode outcomes in X ’s sample space $\mathcal{X}$.

Kolmogorov complexity $K(x)$ is the length of the shortest Turing machine program that prints x , then halts. The shorter an object’s Kolmogorov complexity, the simpler it is and the easier it is to describe it. For example, $K(f)$ for a probability distribution f is large if it is trickily skewed or otherwise “complicated.” Kolmogorov complexity differs between computers by a constant, so we just consider $K(x)$ for a universal Turing machine U .

A cornerstone theorem in Shannon information theory is the Source Coding Theorem, which states that a random variable’s entropy is the lower bound for the average number of bits per word needed by an encoding scheme. We can get arbitrarily close with Huffman codes, for example. Using

Kolmogorov complexity, Shannon's Source Coding Theorem can be stated as

$$H(X) \leq \sum_x f(x)K(x).$$

Armed with this result and a few other definitions, the authors present an important (existing) result:

Theorem 1 *Expected Kolmogorov complexity equals Shannon entropy. Let f be the probability distribution of X . Then*

$$0 \leq \sum_x (f(x)K(x) - H(x)) \leq K(f) + O(1).$$

Whereas $H(x)$ depends on the sample space $\mathcal{X}$, $K(x)$ gives the shortest code word length for x irrespective of its source distribution. In the case that f is simple, $K(f)$ is small, so the shortest general description of x must be close to its entropy, the length needed to describe x in this specific sample space. Because $H(x)$ is already defined given the distribution f , the difference measures how much more information is needed to describe x *universally*, independently of its source distribution. This additional necessary information is at least $K(f)$, the tersest description of the f implicit in $H(x)$.

Alternatively, Cover and Thomas state the theorem like so, using X^n to denote a string of n i.i.d. characters:

$$\mathbb{E} \left[\frac{1}{n} K(X^n | n) \right] \rightarrow H(X).$$

Grünwald and Vitányi's proof uses Shannon's source coding theorem and Solomonoff's concept of the universal distribution, defined $\mathbf{m}(x) = 2^{-K(x)}$, which assigns prior probabilities to all strings based on how simply they can be generated. By contrast, Cover and Thomas's proof of the alternate statement is more classical Shannon-esque, using the Kraft inequality and Jensen's inequality. Both proofs use prefix codes, which allow a Turing machine to continually read a sequence and know where each codeword ends without doubling back (Huffman codes are one such example).

It's an amazing fact that the expected length of the shortest description of a random variable is approximately its entropy. This shortest description is like a universal code that works for *all* probability distributions.

2.2 Mutual Information

In this section, the authors present another existing theorem relating two analogous versions of mutual information.

Beyond quantifying the uncertainty of a random variable X , one might want to quantify how much knowing another random variable Y reduces the uncertainty of X . The *information* in Y about X is

$$I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X).$$

The algorithmic mutual information, which is not rooted in probabilities at all, is a more fundamental idea: it's the quantity of information conveyed by an object (y) about an object (x). More formally, it's

$$I(y : x) = K(x) - K(x | y^*) = K(x) + K(y) - K(x, y)$$

where y^* is the shortest description of y . If y says nothing about x , $K(x)$ and $K(x | y^*)$ are equal because knowing y is useless, that is, $I(y : x) = 0$. The more informative y is, the shorter $K(x | y^*)$

needs to be because y tells us almost everything we need to know about x , so $I(y : x)$ is almost as large as $K(x)$. It can be seen from the definitions that both probabilistic and algorithmic information are symmetric.

Theorem 2 *Expected algorithmic mutual information equals probabilistic mutual information.*

$$I(X; Y) = \sum_x \sum_y f(x, y) I(x : y \mid f) + O(1)$$

Where $f(x, y)$ is given as auxiliary information directly accessible by the Turing machine.

The proof is obtained by separating $f(x, y)$ into marginals as sums over x and y , using the definition to express $I(x : y)$ as $K(x) + K(y) - K(x, y)$, and using Theorem 1 to express the expected Kolmogorov complexities of each marginal as entropies whose differences are exactly the mutual information.

This theorem tells us that the information of the random variable X about Y is the expected amount by which an outcome x reduces the length of the description of y . Thus, rather magically, algorithmic information is a sharpening of the probabilistic notion to individual objects.

2.3 Mutual Information Non-Increase

This section does not present a theorem connecting the two theories, but instead surveys the information non-increase principle's different forms.

An important result in Shannon information theory is the data processing inequality, which says that we cannot gain information by processing the outcome. More formally, for any function T , we have that

$$I(X; Y) \geq I(X; T(Y)).$$

In the algorithmic context, it turns out this also holds true of individual objects. Given x , z , and q being a deterministic function computing z from x^* (the shortest expression for x), we have that

$$I(z : y) < I(x : y) + K(q) + O(1).$$

The information z holds about y is at most the information x holds about y plus the cost of describing the transformation from x to z . We cannot learn more from z about y unless new information from q is added, in which case q itself must be described.

This is true even for randomized functions. The authors show that the probability of a randomized function's output increasing the mutual information by d is $O(2^{-d})$, meaning the randomness of the function cannot reveal more information about the object with a non-negligible probability. The universal distribution is used again for this statement and its proof.

Levin used Kolmogorov complexity and algorithmic mutual information to close a hole in Gödel's incompleteness theorem [3]. Levin describes an infinite string he calls "forbidden information" that has infinite mutual information with *any* completion of a universal function. If we really could make a universal Turing machine halt on every input, it would reveal everything about this mystery string, which contains uncomputable, forbidden information. Even randomized extensions of Peano Arithmetic cannot solve unsolvable problems with non-negligible probability.

2.4 Sufficient Statistics

This section contains a new result that ties classical statistical concepts of sufficiency to an algorithmic analogue. For this we need to define several things:

A sufficient statistic is the tersest way to express what x tells us about the parameters of a model. This is neatly encapsulated in its mutual-information theoretic definition:

$$I(\Theta; X) = I(\Theta; S(X)).$$

An algorithmic sufficient statistic of x is a shortest program for a set $S \ni x$, where $K(x) = K(S) + \log|S| + O(1)$. We say S is *optimal*, meaning x is maximally random with respect to S , and the extra work is only needed to index x in the set using $\log|S|$ bits.

A sequential statistic assigns a string to an “explanation set.” x has some regularity or structure that is encoded in this set, of which x is a random or typical member. It’s a function $S : \{0, 1\}^* \rightarrow 2^{\{0, 1\}^*}$ such that $\forall n, \forall x \in \{0, 1\}^n$, the following hold: (1) $S(x)$ only contains strings of the same length as x ; (2) $x \in S(x)$; and (3) each string is assigned to only one explanation set.

A sequential statistic is sufficient if the program generating $S(x)$ is an algorithmic sufficient statistic for x , i.e. $K(S(x)) + \log|S(x)| \leq K(x) + c$ for some constant c . That statistic is nearly sufficient if we allow the statistic to lose at most a small, constant amount of information.

Theorem 3 *Let S be a sequential sufficient statistic. If $\forall f_\theta$ in a model class, $K(f_\theta)$ is finite, then S is a nearly-sufficient statistic for f_θ with respect to the uniform distribution.*¹

The proof uses an equivalent condition for sufficiency, which is that if there exists a function $g : \mathcal{X} \times \mathcal{S} \rightarrow \mathcal{R}$ such that $g(x | s) = f_\theta(x | s)$ for all $x \in \mathcal{X}, s \in \mathcal{S}$, and probability mass functions f parameterized by $\theta \in \Theta$, then S is sufficient. The authors prove this condition using g as the uniform distribution. In other words, given $S(x)$, f_θ will assign x a probability very close to that of a uniform distribution. Because x is maximally random in its optimal set and f is given the set, the probability that it is any of the items in the set is about the same.

The proof also uses the optimality of Shannon-Fano codes and the information inequality, which essentially states that the expected number of additional bits incurred by non-optimal codes is positive for imperfect approximations of f_θ ². Non-optimal codes require more bits, and this results in a series of inequalities that prove the near-sufficiency of S .

This theorem tells us that algorithmic sufficient statistics are probabilistic nearly-sufficient statistics with respect to *all* model families. Once again, we see the algorithmic notion generalize its probabilistic equivalent; reassuringly, we arrive at the same principle for a sufficient statistic, whether defined in terms of probabilistic information or description length. Information about data points or objects is maximally condensed in sets of uniform probability—precisely when entropy is maximized.

2.5 Rate Distortion

In this section, the authors present a new result relating the Kolmogorov structure function and the distortion of a random variable.

In this setting, the number of bits used to describe a random variable or object is limited. If one is restricted to R bits and the set of outcomes $\mathcal{X}$ has cardinality $|\mathcal{X}| > 2^R$, or the object has $K(x) > R$, information is necessarily lost in the compression. Because there no perfect decoding

¹Note that “nearly sufficient” is as good as we can get. This is because definitions of $K(f)$ are up to a constant, but the definition of a sufficient statistic requires exact equalities, hence the need for the more relaxed definition.

²The information inequality may be more familiar to the reader in this form: *The Kullback-Leibler divergence between two distributions f and g is always positive, with equality iff $f \equiv g$.*

function, we measure how “good” this decoder is using a distortion function $d: \mathcal{X} \times \mathcal{Y} \rightarrow [0, \infty)$, or $\frac{1}{n} \sum_{i=1}^n d(x_i, y_i)$ for a sequence of random variables. Common distortion functions include the Hamming or Euclidean distances.

To state their theorem, the authors define the minimum expected distortion for decoding functions $Z_n: \mathcal{X}^n \rightarrow \mathcal{Y}^n$ using at most R bits per character:

$$D_n^*(R) = \min_{Z_n: |\mathcal{Z}_n| \leq 2^{nR}} \mathbb{E}[d(X^n, Z_n)].$$

The Kolmogorov structure function is defined

$$h_x(R) = \min_S \{\log |S| : S \ni x, K(S) \leq R\}$$

and can be understood as the size of the smallest model S for the data x . h_x can also be thought of as the maximum likelihood estimator on models constrained in complexity by R . By selecting the smallest model, h_x exhibits a bias toward the least complex explanation for the data. In this way, Kolmogorov’s structure function formalizes Occam’s Razor.

Theorem 4 *Expected structure function equals rate distortion. Let d be a distortion measure and R be the limit on bits per character. Suppose X takes values $\{0, 1\}^n$ according to $f(x)$. Then, for a string of m i.i.d. outcomes $\bar{x} = x_1 x_2 \cdots x_m$,*

$$\mathbb{E} \frac{1}{m} h_{\bar{x}}(mR + K(f, d, m, R) + O(\log(n))) \leq D_m^*(R) \leq \mathbb{E} \frac{1}{m} h_{\bar{x}}(mR).$$

The proof uses distortion spheres to create coverings of $\{0, 1\}^n$. The idea is that if the set of all possible outcomes lies within the distortion radii of spheres centered at outputs of the decoding function, then all $x \in \{0, 1\}^n$ can be reached by the decoding function within a limited error.

The Kolmogorov structure function’s combinatorial approach to data fitting is not probabilistic at all; it is concerned entirely with finding sets within which the x ’s are “typical” and can thus be described by a simple log index. In this way, the structure function can *always* determine the best-fitting model at every model-complexity level and thus find the best fit, and not only with high probability. It is thus a rather beautiful result that taking its expectation with respect to the output strings’ probabilities yields the minimal achievable distortion for any decoding function.

3 Conclusion

We have seen in this paper that algorithmic information theory can generalize concepts in Shannon information theory; whereas Shannon’s theory describes the information of random variables with respect to their probability distributions, algorithmic information theory allows us to describe individual objects universally.

At first glance, it tracks that two fields concerned with measuring meaningful information content are so closely related. But it’s quite astonishing that two completely different fields, driven by very different motivations (Shannon wanted to send messages concisely and Solomonoff wanted to make Artificial General Intelligence) could describe the same thing.

Kolmogorov complexity is more general than entropy, but it is nebulous and less practically useful (it is literally incomputable). Cover and Thomas summarize its significance like so: Kolmogorov complexity is a conceptual precursor to entropy; it formalizes a way to think more clearly about computer science and communication theory.

References

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